D14 Capimagics

Title Project: Neuromorphic standard cells with gLayout

Discord name	Affiliation	Experience	Role
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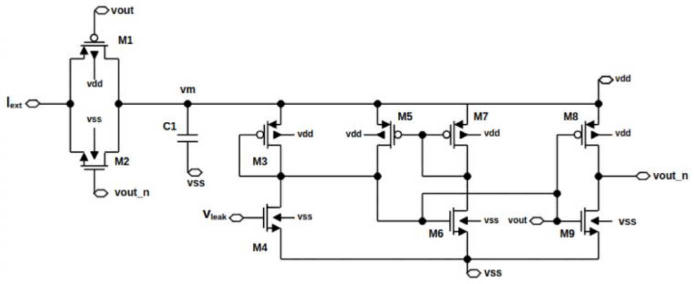
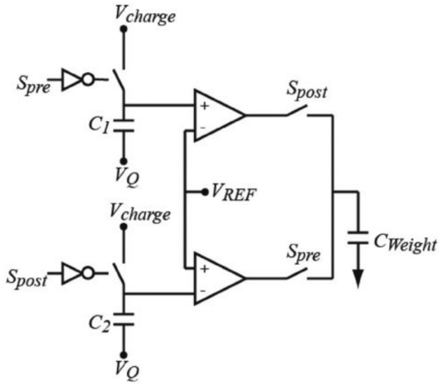
Project information (1-2 pages)

- Goal (2 sentences)
  - Implement in gf180 a set of neuromorphic standard cells using glayout
    - Neuron
    - STDP synapse
    - Analog-Spiking Encoding/Decoding circuits
  - Automatize the assemble a small spiking neural network with glayout
    - Create an small, reproducible SNN as testbench
- Implement the necessary code using glayout/librelane to do the hardening and the physical synthesis for tapeout

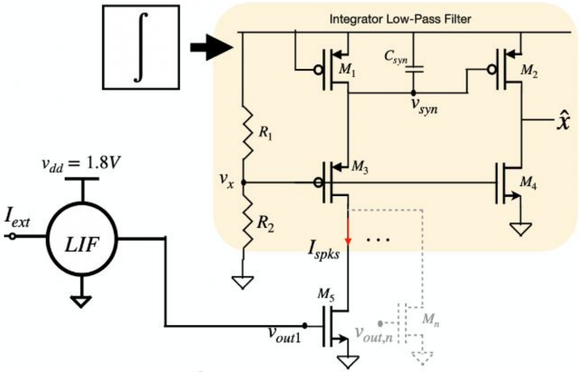
Project information (1-2 pages)



LIF Neuron



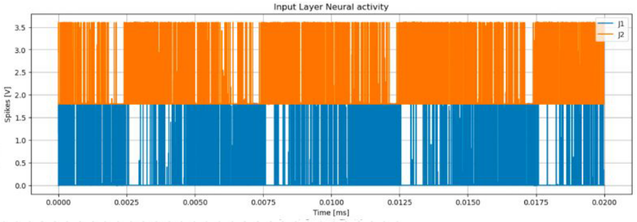
STDP Synapse (non-rram)

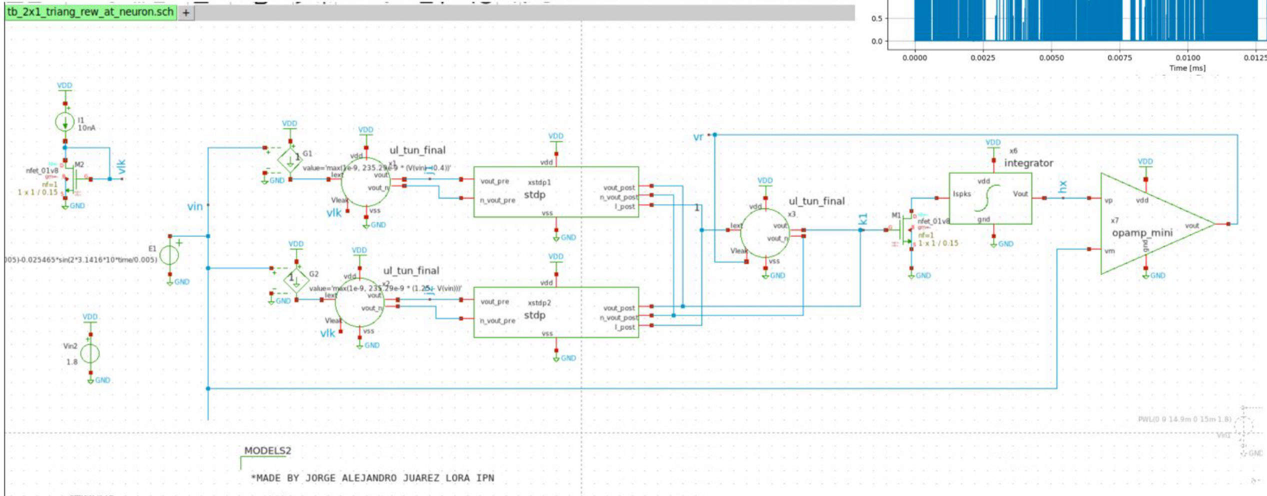


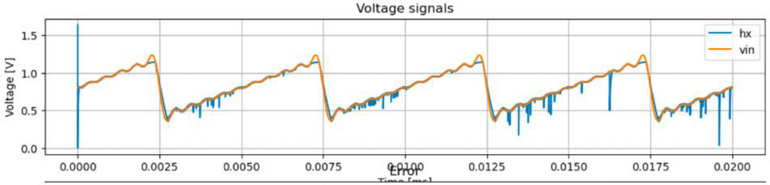
Spike integrator

Application

Construct a network (layout, gf180) that learns to reconstruct a signal to be learnt as test.







Team Background

Academic Experience
1 Dr, 1 phD student, 1 Master Student, 1 Bachelor
Work Experience
Teaching experience (Alex), Electronic Design Engineering (Katia, Jalton)

Questions

- Regarding analog circuits in gf180, how many pins would we have available for tapeout?
- Is it possible to create a generator for gLayout, for SNNs?
- Should a gLayout + Librelane implementation would be better, or it can be done with plain glayout?

References

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